

1 Development Suite MES-Cockpit - XML File Interface

The defined XML file interface also used by default can be implemented to integrate customer-specific basic KPIs.

The requirements for using the XML file interface are:

- The structure described here has been adhered to
- Data is transferred for existing data types and/or new data types have been created in MES-Cockpit as described in the document entitled MESC_DevelopmentSuite.pdf.
- Meta data/master data of objects are included
- Data to be transferred has to be filed in a new directory on the MES-Cockpit server. The directory name starts with the object name plus an underscore character "_". e.g. Workplace_

The defined storage locations can be found in the configuration file MESC_DataLocations.txt (c:\ProgramData\QlikTech\Documents\conf\)

1.1 General

The MESC_Loader.qvw application processes serialized data in XML format. Several types are differentiated. Different types of data are stored in separate directories.

1.2 Master data

Master data is stored in a separate file for each site according to the following pattern:

```
<?xml version="1.0"?>

<MescData xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">

  <Meta>

    <Plant>Werk4</Plant>

    <ObjectType>Operation</ObjectType>

    <DataType>masterdata</DataType>

    <CreationTime>2014-09-09T07:48:44.9645867+02:00</CreationTime>

    <ValidFrom>0001-01-01T00:00:00</ValidFrom>

    <ValidTo>0001-01-01T00:00:00</ValidTo>

  </Meta>
```

```

<Content>

  <MasterData>

    <Id>Id0</Id>

  <Info>

    <Parameter Type="decimal" Name="dec1" Value="8000.000000" />

    <Parameter Type="string" Name="text1" Value="value" />

    <Parameter Type="number" Name="num1" Value="18000" />

    <Parameter Type="datetime" Name="time1" Value="2004-06-18 06:20:16" />

  </Info>

</MasterData>

</Content>

```

General information on data is filed within the meta element. The internal site ID is indicated in the plant element. With master data the DataType element is set to masterdata. The object type is defined in the ObjectType element.

Data objects are stored in MasterData elements (as many as required for each XML). Each object is identified by a distinct ID. As many parameters as you like can be indicated for each object. Each parameter has a specific type (decimal, string, number, datetime), a name and a value that is formatted accordingly. Number parameters are integers; decimal parameters are decimals with 6 decimal places (dots are used as decimal separators) and time stamps are specified according to the ISO format.

1.3 Basic KPIs

Basic KPIs are determined and stored for each shift.

```

<?xml version="1.0"?>

<MescData xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">

  <Meta>

    <Plant>Werk2</Plant>

    <ObjectType>Operation </ObjectType>

    <DataType>keyfiguredata</DataType>

    <CreationTime>2014-08-29T09:08:26.3090441+02:00</CreationTime>

```

```
<ValidFrom>2012-01-11T00:00:00</ValidFrom>

<ValidTo>2012-01-11T00:00:00</ValidTo>

</Meta>

<Content>

  <Block ObjectId="Id1" RefDate="2012-01-11" ShiftNo="1">

    <KeyFigures>

      <Entry Key="kf1" Value="1.000000" />

      <Entry Key="kf2" Value="1.234567" />

    </KeyFigures>

  </Block>

</Content>

</MescData>
```

Similar to master data, general information is stored in the meta element and the data itself is filed in the content element.

Data is stored in as many sections/block elements as required. Each block is distinctly identified by an object ID (ObjectId), the shift date (RefDate) stated according to the ISO format and the shift number (ShiftNo). KPIs are stored as entries in the block. KPIs are always decimals with 6 decimal places (dots are used as decimal separators).